

US Media coverage of China:

A Comparative Study with Keyword-Assisted Topic Model

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Introduction

U.S. media coverage of China has consistently carried a somewhat negative connotation, often framed through the lens of geopolitical rivalry and ideological tension. Yang and Liu (2012) found that between 1992 and 2006, U.S. print media frequently portrayed China as a threat, whether military, economic, political, or ideological, based on a qualitative content analysis of major newspapers. Building on their work, this study seeks to explore how U.S. media currently portrays China and to identify key topics emphasized in recent publications and see if these topic goes beyond military, economic, ideology, and politics.

To ensure objectivity, I will compare the coverage of China with that of Japan. Japan is one of the world's largest economies, a major trading partner in the Asia-Pacific region, and a close U.S. ally, frequently depicted in a more favorable light. By comparing media portrayals of China and Japan, I aim to reveal both distinct differences and shared thematic patterns.

The primary objective of this research is to identify and frame the dominant topics in U.S. media coverage of China. As noted by Chang et al. (2009), traditional topic models such as LDA and STM often yield vague or ambiguous themes, sometimes described as “reading tea leaves”, because they cluster similar words without assigning clear semantic meaning.

To address this limitation, I adopt a semi-supervised approach using the Keyword-Assisted Topic Model (KeyATM), first proposed by Eshima and Sasaki (2024). This method allows for the manual assignment of topic labels based on a set of predefined keywords, improving interpretability while maintaining flexibility in the modeling process. My research

finds that U.S. media tend to portray China through a more socio-political lens, with coverage focusing on topics such as elections, domestic policies, foreign relations, Asia-Pacific geopolitical tensions, technology, and trade. In contrast, media coverage of Japan leans more toward a geo-cultural perspective. While it also includes some of the socio-political themes mentioned above, there is a noticeably stronger emphasis on entertainment in the coverage of Japan.

Data Collection and Methods.

All data for this research were collected using the New York Times Article Search API. I retrieved all articles published between January 1 and December 31, 2024, that mention “China” or “Japan” in the headline, abstract, or main body. My primary interest is in how U.S. news analyzes China, but I also aim to capture the broader contexts in which China is mentioned. To avoid missing relevant content, especially in more recent publications, I chose not to limit the search to headlines alone. While scanning full articles increases the chance of including irrelevant mentions, it allows for a more comprehensive dataset.

Due to the API's limitations in accessing full article texts, I combined the headline and abstract of each article to form the unit of analysis. Data was queried daily throughout the year, resulting in a total of 4,241 articles related to China and 1,812 articles related to Japan.

The word count statistics for the combined headline and abstract texts are as follows:

- **China-related articles:** median word count = 4, maximum = 60, minimum = 31
- **Japan-related articles:** median word count = 4, maximum = 53, minimum = 30

For text preprocessing, I utilized the [quanteda](#) package in R to remove punctuation, numbers, and English stopwords. Additionally, I eliminated commonly occurring but

semantically insignificant words such as “briefing” and weekday names. Since the goal is to fit a keyword-assisted topic model, which requires complete words for semi-supervised learning, I did not apply stemming to the corpus. To preserve the corpus's integrity and ensure that all words have the potential to contribute to exploratory data analysis and keyword selection, I did not exclude words based on their frequency, whether too common or too rare.

Analysis

To achieve my research objectives, I began by generating word clouds and employing Structural Topic Modeling (STM) as part of an exploratory analysis for keyword selection. Additionally, I plotted the distribution of the original topics, which are the generic labels assigned by The New York Times, to understand the broader categories each news article falls into. These steps aimed to uncover latent themes within the corpus before assigning manually selected labels.

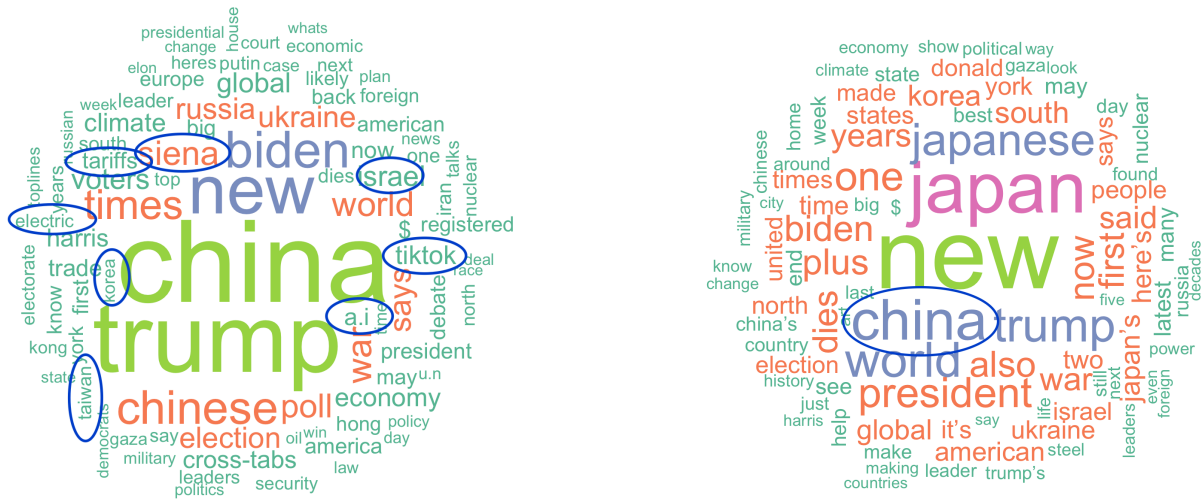
For STM, I utilized the built-in `searchK` function to determine the optimal number of topics (k) by balancing held-out likelihood, semantic coherence, residuals, and lower bounds. Based on the insights from the word clouds and STM results, I selected around 5 keywords on average for each article corresponding to each manually assigned topic. Finally, I conducted a covariate analysis to identify the driving factors behind topic formation.

Results

Word Clouds

Below is the word cloud for China (left) and Japan (right). As expected, the most frequently mentioned terms are *China* and *Japan*. Notably, *Trump*, the current president of the

United States, also appears prominently. This suggests that both countries are often discussed in the U.S. presidential election campaign.

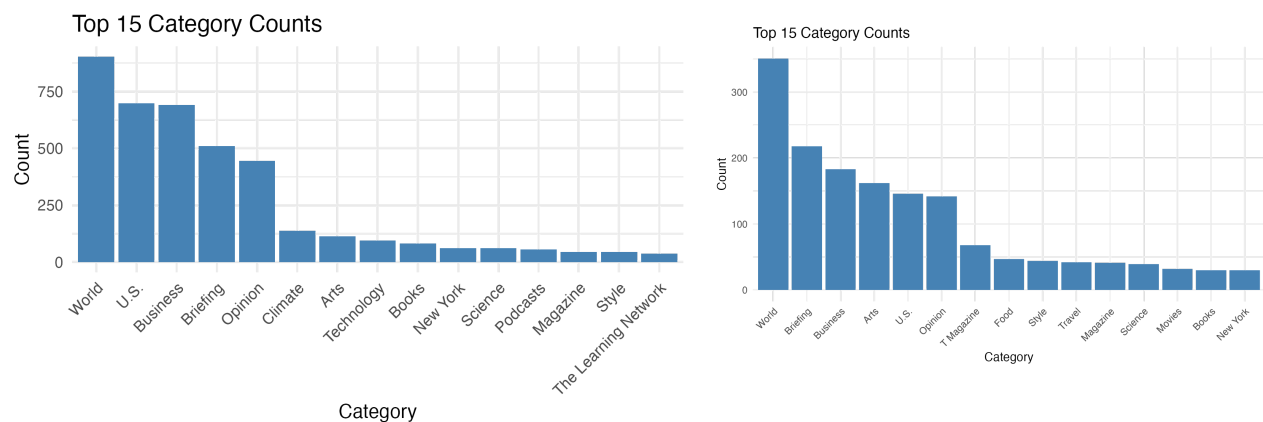


In the Chinese word cloud, several prominent themes emerge. Technology-related terms such as *TikTok*, *AI*, and *electric* (often appearing in the context of “electric cars”) are notably frequent. Election-related keywords like *Siena* and *poll* also stand out. Furthermore, the word cloud highlights foreign policy and Asia-Pacific terms, including *Taiwan*, *Korea*, *Ukraine*, and *Israel*. Additionally, U.S. domestic political figures such as *Biden*, *Kamala*, *Harris*, and *Donald*, are prominently featured, reflecting discussions on partisanship and political dynamics.

In the Japanese word cloud, *China* emerges as one of the most prominent keywords, indicating its centrality in the discourse. Other frequently appearing terms align with themes such as elections and presidential campaigns, domestic politics, international relations, and Asia-Pacific affairs. This pattern suggests that Japanese media coverage often intertwines discussions of China with broader political and regional contexts.

True Topic Distribution

Based on the topic labels assigned by The New York Times, the three most prevalent categories for China-related articles (left) are World, “U.S.,” and “Business,” followed closely by “Briefing” and “Opinion.” In contrast, while “World” and “Business” also dominate the Japan-related articles, “Arts” emerges as the fourth most frequent category. Additionally, Japan's coverage (right) includes more culturally oriented topics such as “Food,” “Travel,” and “Style,” which are scarcely mentioned in the context of China. This distribution suggests that U.S. media coverage of Japan tends to emphasize cultural and lifestyle aspects, whereas reporting on China is more closely tied to political and economic affairs.

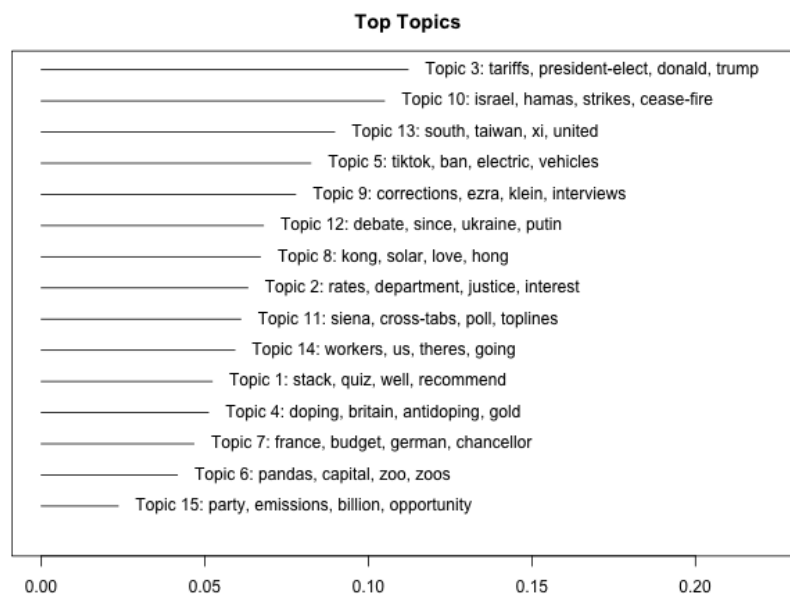


STM

For the Structural Topic Modeling (STM) analysis, I selected 15 topics for the China corpus and 20 topics for the Japan corpus. These selections were informed by diagnostic metrics from the `searchK` function in the `stm` package to achieve low residual dispersion and high semantic coherence, both of which are indicators of well-defined and interpretable topics.

Although the held-out likelihoods for both models were relatively low, suggesting limited predictive performance on unseen data, this metric was not a primary concern for this study.

Since the focus is on uncovering latent themes within the existing corpus rather than predicting new content, the held-out likelihood was not emphasized in the model selection process.

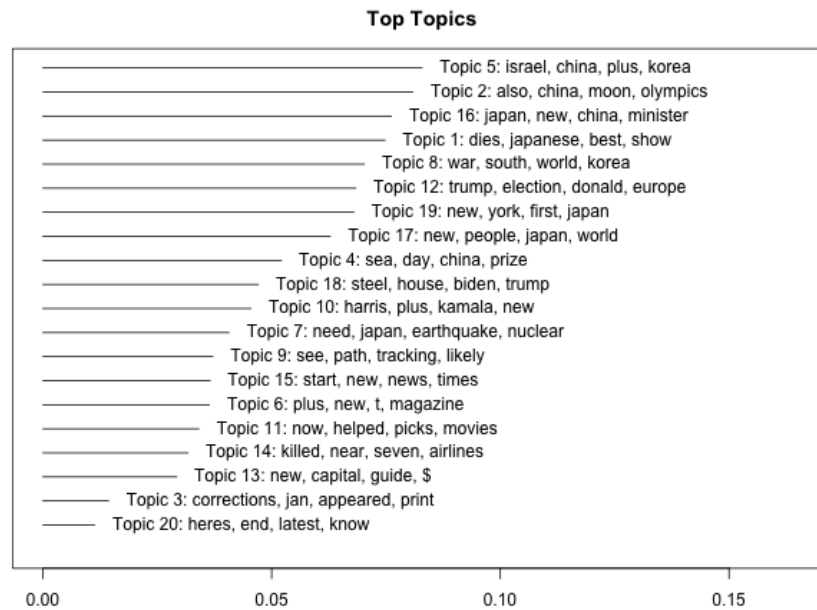


The visualization above shows the topic distribution and associated keywords for articles related to China. Based on the topic modeling results, ordered from most to least frequent, Topic 3 appears most frequently in the China-related corpus. This topic reflects themes of trade policy and tariffs that former President Trump was expected to pursue if re-elected and later confirmed. Topics 10, 13, 9, 12, and 8 broadly cover foreign relations and general geopolitical tensions in the Asia-Pacific region. Topic 11 focuses on the U.S. election, while Topic 4 pertains to the 2024 Paris Olympics.

The visualization below shows the topic distribution and associated keywords for articles related to Japan. In the case of Japan, the topic distribution is more diffuse, making it difficult to assign each topic to a distinct category. However, Topics 5, 16, and 8, which are among the most frequently occurring, generally relate to Asia-Pacific political dynamics and foreign affairs.

Topics 12, 18, and 10 are associated with the U.S. presidential campaign. Additionally, themes of

entertainment can be inferred from Topics 1, 6, and 11, which include terms such as *best*, *show* (referring to *Shogun*, as confirmed by contextual analysis), *magazine*, and *movies*. Other themes include the Olympics, represented by Topic 2, and Natural Disasters, represented by Topic 7.



Keyword Selection

Based on the qualitative analysis presented above, I selected the following keywords for China.

APAC: taiwan, korea, hong, kong, japan

Election: election, vote, siena, poll

Economy: economy, market, stock

Foreign_Policy: putin, russia, ukraine, iran, israel, gaza

Olympics: olympics, tracking,

Technology: tiktok, a.i, cars, electric, vehicles, tech

Trade: trade, tariffs, imports, exports

US_Politics: trump, republican, vance, electorate, democrats, biden, harris

Japan and China share many common themes. The following keywords were selected for Japan.

APAC: china, taiwan, korea

Entertainment: show, movies, shogun, art, tv, fashion

Election: election, vote,

Economy: economy, bill, market, stock,

Foreign_Policy: putin, ukraine, iran, israel, gaza,

Olympics: olympics, tracking,

Technology: a.i, tech

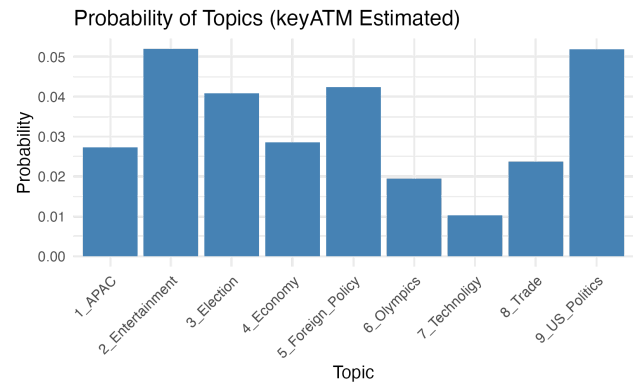
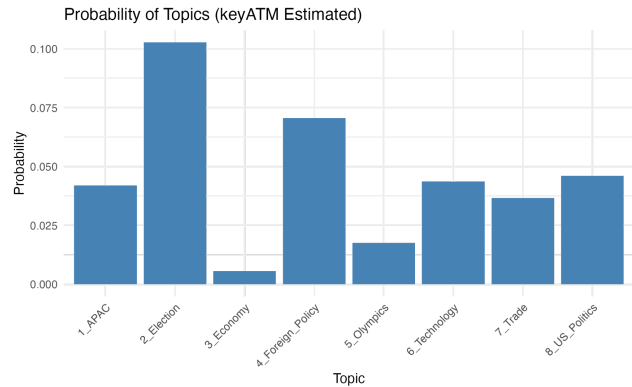
Trade: trade, tariffs,

US_Politics: trump, republican, vance, democrats, biden, kamala, harris

For both China and Japan, I selected topics that appear in both countries to examine how common themes are distributed across the corpus. As previously noted through the word cloud, New York Times labels, and STM results, I also included a separate topic on Entertainment, which does not appear in the China-related corpus. Keywords for each topic were selected based on patterns observed in the word clouds and STM outputs. It is worth noting that some keywords under themes like APAC and Technology were adapted to better reflect the Japanese context. For instance, media coverage in Japan related to the Asia-Pacific region tends to focus more on *China* than on *Hong Kong*. Similarly, Japanese technology-related themes are less focused on *electric vehicles*. The overall distribution of keywords across the corpus is shown in Figures 3 and 4 in the appendix.

KeyATM Results

From the keyword-assisted topic model (see Figures 5 and 6 in the Appendix), we can see that the model effectively captured most of the manually selected keywords. In addition, it identified many additional words that align with the intended themes but were not originally selected as keywords (highlighted in orange). Below is the distribution of KeyATM themes for both China (left) and Japan (right).



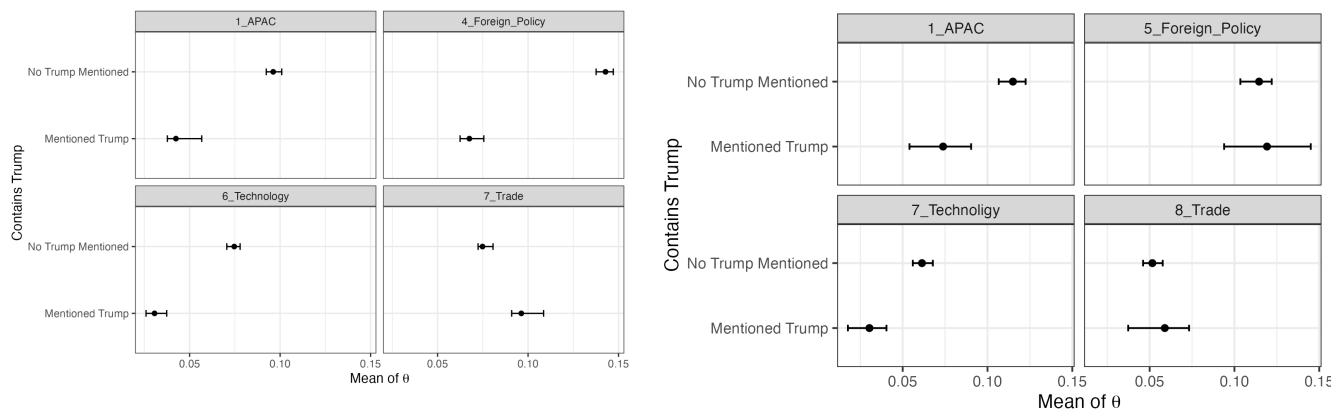
The topic distribution reveals that issues related to China are heavily concentrated in themes such as Election and Foreign Policy. This suggests that media coverage of China tends to emphasize socio-political dimensions, often in connection with U.S. domestic policies. In contrast, topics related to Japan are more evenly distributed, with a notable emphasis on entertainment.

Covariate analysis: A difference and difference approach.

To explore potential factors driving topic formation for China and Japan, I created a binary variable called `contains_trump`, since *Trump* is one of the most frequently occurring words in both corpora and refers to the current president of the United States. I aim to examine whether the presence of *Trump* in the headline or abstract increases or decreases the likelihood of an article being classified under certain topics.

As expected, when Trump is mentioned, the likelihood of being classified as *US_Politics* increases for both China and Japan. As we can see in figures 7 and 8 in the appendix. However, when it comes to other topics, we see some differences.

The figure below shows the covariate analysis for both China (left) and Japan (right). As we can see from the China results, mentions of Trump decrease the likelihood of an article being classified under themes such as Asia-Pacific, Foreign Policy, and Technology. In contrast, the probability of classification under the Trade theme increases significantly. This suggests that, based on 2024 news coverage, China is more frequently framed as a target of President Trump's trade policies compared to Japan. When Trump is mentioned, the articles are still classified as related to trade, but the framing tends to reflect U.S. political discourse or takes on a confrontational tone.



In contrast, for Japan, we do not observe a similar increase in the likelihood of trade-related classification when Trump is mentioned. Despite Japan being one of the United States' largest trading partners and a major car exporter, it is not as frequently associated with discussions around trade deficits or tariff disputes in the U.S. media.

Conclusion

Through our analysis using word clouds, topic distribution, Structural Topic Modeling (STM), and Keyword-Assisted Topic Modeling (KeyATM), we find that U.S. media coverage of China primarily focuses on elections, U.S. domestic policy, foreign relations, Asia-Pacific

geopolitical tensions, technology, and trade. Notably, China is frequently discussed in the context of presidential election campaigns and domestic political discourse. This suggests that U.S. media tend to portray China through a sociopolitical lens, particularly about trade, reinforcing a narrative often reserved for geopolitical “rivals.”

In contrast, while themes such as elections, U.S. politics, and foreign relations also appear in coverage of Japan, our analysis reveals a strong emphasis on Japan’s sociocultural dimension. The Entertainment theme consistently emerges across all modeling approaches, aligning with the common portrayal of Japan as a cultural “ally” in U.S. media.

For future work, we could expand our analysis to include media coverage from multiple outlets to assess whether topic composition varies by political orientation. Additionally, extending the time frame to 5-10 years may reveal longitudinal trends in media framing.

This research contributes to the growing field of cross-cultural media studies and offers insight into how textual analysis can illuminate cultural dynamics and geopolitical tensions. Further exploration could include sentiment analysis within specific cultural contexts or the use of tuned word embeddings to understand how key terms are framed across different narratives.

Reference

- Chang, Jonathan, Jordan Boyd-Graber, Sean Gerrish, Chong Wang, and David M. Blei. 2009. "Reading Tea Leaves: How Humans Interpret Topic Models." In Proceedings of the 22nd International Conference on Neural Information Processing Systems (NIPS '09). 288–96.
- Eshima, S., Imai, K., & Sasaki, T. (2024). Keyword-assisted topic models. *American Journal of Political Science*, 68(2), 730-750.
- Yang, Y. E., & Liu, X. (2012). The 'China threat' through the lens of US print media: 1992–2006. *Journal of Contemporary China*, 21(76), 695-711.

Appendix 1

Figure 1: K performance for STM for China

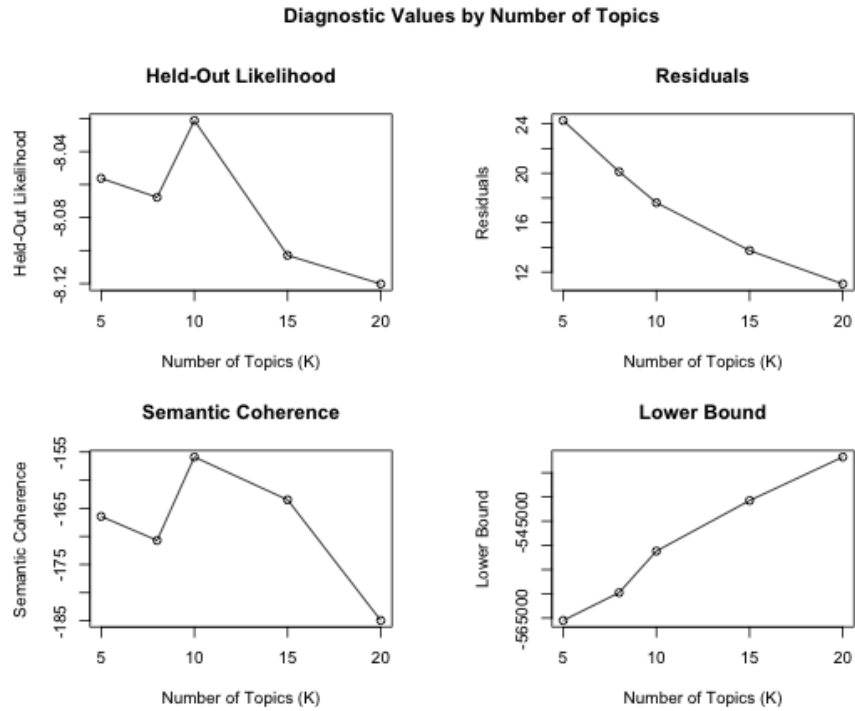


Figure 2: K performance for STM for Japan

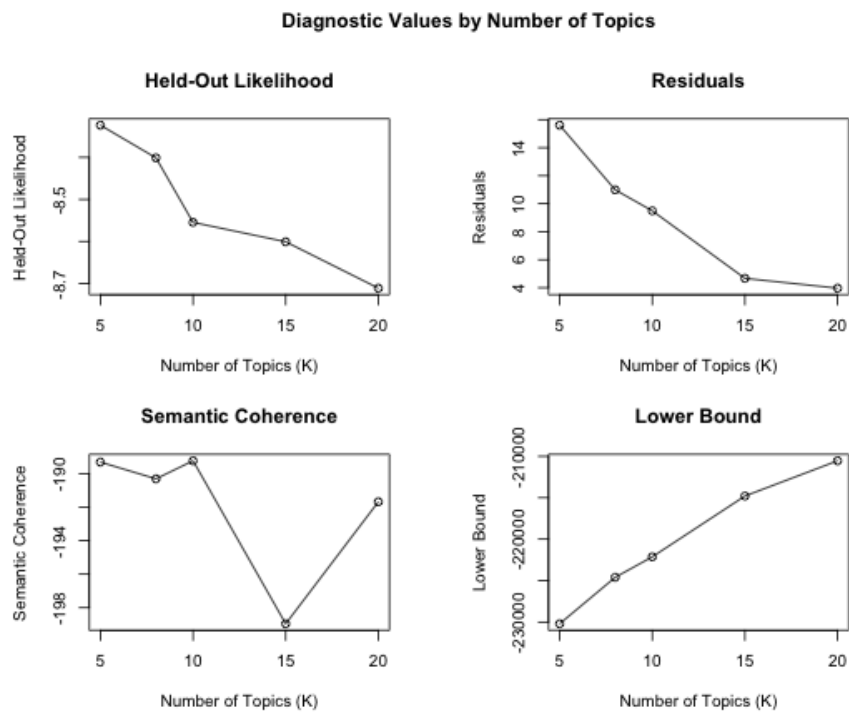


Figure 3: Keyword distribution for China
(Separated due to difference in y axis)

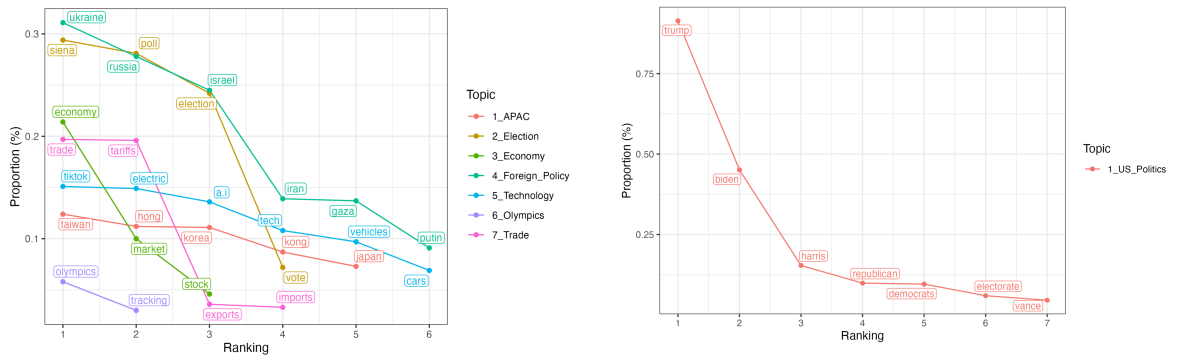


Figure 4: Keyword distribution for Japan
(Separated due to difference in y axis)

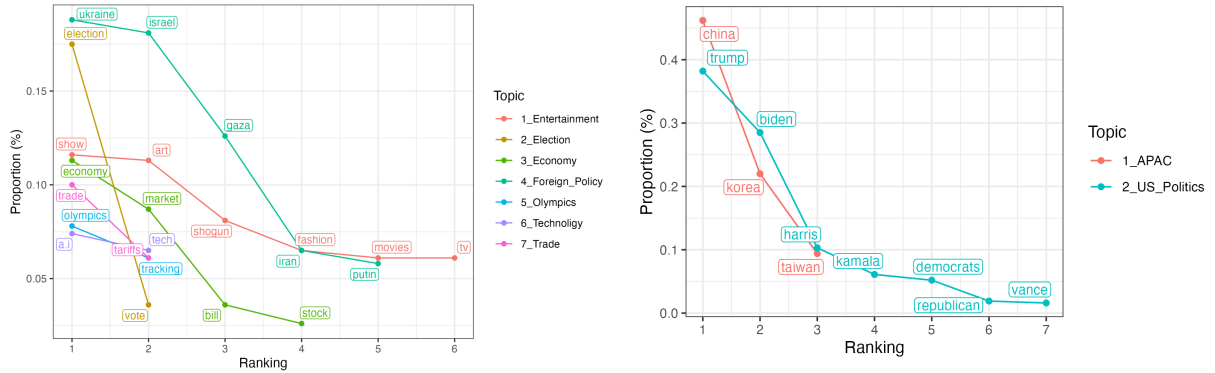


Figure 5: KeyATM results for China

Keyword Distribution for China					
1_APAC	2_Election	3_Economy	4_Foreign_Policy	5_Technology	6_Trade
taiwan [✓]	times	economy [✓]	ukraine [✓]	electric [✓]	trade [✓]
hong [✓]	siena [✓]	new	russia [✓]	a.i [✓]	tariffs [✓]
korea [✓]	poll [✓]	economic	israel [✓]	tiktok [✓]	new
south	voters	global	war	company	trump [7]
climate	new	market [✓]	also	vehicles [✓]	global
law	york	growth	president	tech [✓]	states
kong [✓]	college	may	iran [✓]	new	president
world	results	investors	gaza [✓]	billion	united
japan [✓]	likely	prices	nuclear	cars [✓]	donald
power	conducted	leaders	military	elon	american
7_US_Politics	Other_1	Other_2	Other_3	Other_4	
trump [✓]	new	new	years	one	
biden [✓]	klein	world	new	world	
president	show	heres	questions	said	
donald	ezra	people	one	american	
former	york	paris	help	became	
election [2]	week	also	said	new	
harris [✓]	times	moon	last	real	
policy	reading	may	time	least	
presidential	central	first	war	intelligence	
foreign	years	election [2]	found	just	

Figure 6: KeyATM results for Japan

Japanese Keyword						
1_APAC	2_Entertainment	3_Election	4_Economy	5_Foreign_Policy	6_Olympics	7_Technology
china [✓]	new	election [✓]	japan	ukraine [✓]	japan	moon
korea [✓]	show [✓]	dies	economy [✓]	israel [✓]	times	first
south	now	first	markets	war	tracking [✓]	japan
japan	series	street	market [✓]	plus	see	a.i [✓]
north	art [✓]	women	new	new	likely	airlines
taiwan [✓]	shogun [✓]	wall	world	also	path	restaurant
two	time	vote [✓]	investors	gaza [✓]	storm	years
president	first	plus	interest	united	tropical	news
minister	movies [✓]	adams	japans	states	medal	family
leader	fashion [✓]	power	stocks	global	typhoon	japanese
8_Trade	9_US_Politics	Other_1	Other_2	Other_3	Other_4	
steel	trump [✓]	dies	new	nuclear	new	
trade [✓]	president	japanese	japan	years	york	
japanese	biden [✓]	now	years	last	times	
company	donald	one	york	dies	plus	
china [1]	trumps	country	city	best	week	
japan	new	war	made	hong	magazine	
industry	also	world	japanese	stories	hot	
companies	heres	found	people	one	found	
tariffs [✓]	harris [✓]	home	artists	japanese	see	
billion	policy	game	world	said	recommendations	

Figure 7: Covariates Analysis for China in US_Politics

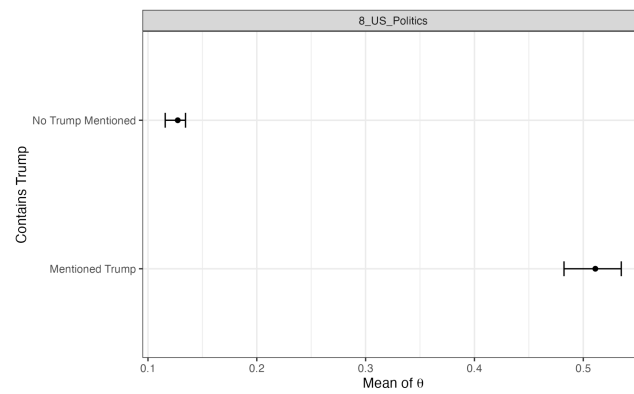


Figure 8: Covariates Analysis for Japan in US_Politics

